

Hao Zhou

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<https://hao23zh.github.io>

Research Interests

Numerical analysis; geometric partial differential equations; finite element methods; structure-preserving numerical methods.

Education

Peking University, Beijing, China

Ph.D. in Mathematics

2023 – Present

Thesis: *Finite Element Methods for Fourth-Order Elliptic PDEs on Surfaces* (proposal stage)

Adviser: Prof. [Shounan Wu](#)

B.Sc. in Mathematics

2019 – 2023

Thesis: *Local Discontinuous Galerkin Method for Fractional Laplacian*

Adviser: Prof. Shounan Wu

Publications

[Google Scholar Profile](#)

- [1] *A stabilized nonconforming finite element method for the surface biharmonic problem*. S. Wu, and H. Zhou; SIAM Journal on Numerical Analysis (2025). [DOI](#)
- [2] *A monotone discretization for the fractional obstacle problem and its improved policy iteration*. R. Han, S. Wu, and H. Zhou; Fractal and Fractional (2023). [DOI](#)

Preprints

- [1] *Local discontinuous Galerkin method for the integral fractional Laplacian*, R. Han, S. Wu, and H. Zhou; arXiv preprint (2025). [DOI](#)

Honors and Awards

- Young Science and Technology Talent Cultivation Project (Doctoral Program), China Association for Science and Technology (CAST), 2025
- Graduate Student Research Showcase, 1st Prize, Peking University, 2025
- Excellence Program in Applied Mathematics and Statistics, Peking University, 2023